



Electronic I

Lecture 2

Semiconductor Materials: Periodic table of elements;

Element semiconductors; and PN Junction

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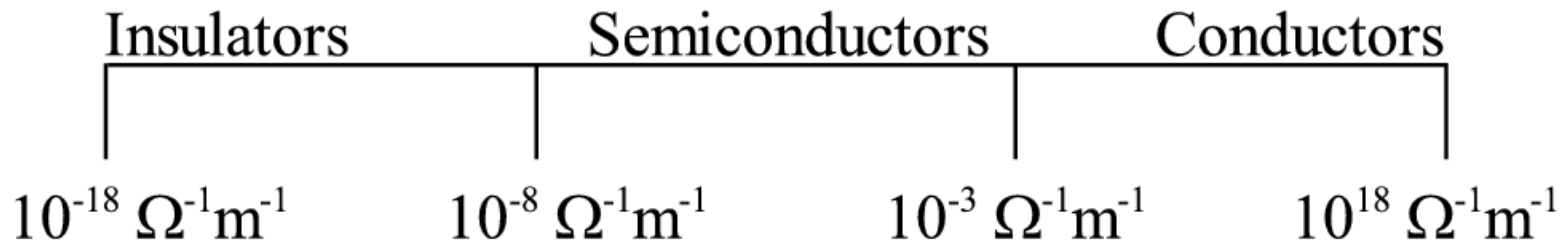
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Introduction

Semiconductors are materials whose electrical properties lie between Conductors and Insulators.

Ex : Silicon and Germanium

Difference in conductivity



Increasing conductivity

- The conductivity of the semiconductor material increases when the temperature increases.
- This is because the application of heat makes it possible for some electrons in the valence band to move to the conduction band.
- Obviously the more heat applied the higher the number of electrons that can gain the required energy to make the conduction band transition and become available as charge carriers.
- This is how temperature affects the carrier concentration.
- Another way to increase the number of charge carriers is to add them in from an external source.
- Doping or implant is the term given to a process whereby one element is injected with atoms of another element in order to change its properties.
- Semiconductors (Si or Ge) are typically doped with elements such as Boron, Arsenic and Phosphorous to change and enhance their electrical properties.

Semiconductor Materials

- **Elemental semiconductors** – Si and Ge (column IV of periodic table) –compose of single species of atoms
- **Compound semiconductors** – combinations of atoms of column III and column V and some atoms from column II and VI.
(combination of two atoms results in binary compounds)
- There are also three-element (ternary) compounds (GaAsP) and four-elements (quaternary) compounds such as InGaAsP.

(a)	II	III	IV	V	VI
		B	C		
		Al	Si	P	S
	Zn	Ga	Ge	As	Se
	Cd	In		Sb	Te

(b)	Elemental	IV compounds	Binary III-V compounds	Binary II-VI compounds
	Si	SiC	AlP	ZnS
	Ge	SiGe	AlAs	ZnSe
			AlSb	ZnTe
			GaP	CdS
			GaAs	CdSe
			GaSb	CdTe
			InP	
			InAs	
			InSb	

Periodic Table of the Elements

[illegible]

Semiconductor materials

* Lanthanide Series
+ Actinide Series

58 Ce	59 Pr	60 Nd	61 Pm	62 Sm	63 Eu	64 Gd	65 Tb	66 Dy	67 Ho	68 Er	69 Tm	70 Yb	71 Lu
90 Th	91 Pa	92 U	93 Np	94 Pu	95 Am	96 Cm	97 Bk	98 Cf	99 Es	100 Fm	101 Md	102 No	103 Lr

Semiconductor Materials

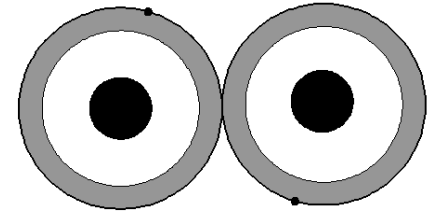
- The wide variety of electronic and optical properties of these semiconductors provides the device engineer with great flexibility in the design of electronic and opto-electronic functions.
- **Ge** was widely used in the early days of semiconductor development for transistors and diodes.
- **Si** is now used for the majority of rectifiers, transistors and integrated circuits.
- **Compounds** are widely used in high-speed devices and devices requiring the emission or absorption of light.
- The electronic and optical properties of semiconductors are strongly affected by impurities, which may be added in precisely controlled amounts (e.g. an impurity concentration of one part per million can change a sample of Si from a poor conductor to a good conductor of electric current). This process called doping.

Possible Semiconductor Materials

Carbon	C	6	1. Very Expensive 2. Band Gap Large: 6eV 3. Difficult to produce without high contamination
Silicon	Si	14	1. Cheap 2. Ultra High Purity 3. Oxide is amazingly perfect for IC applications
Germanium	Ge	32	1. High Mobility 2. High Purity Material 3. Oxide is porous to water/hydrogen (problematic)
Tin	Sn	50	1. Only “White Tin” is semiconductor 2. Converts to metallic form under moderate heat
Lead	Pb	82	1. Only “White Lead” is semiconductor 2. Converts to metallic form under moderate heat

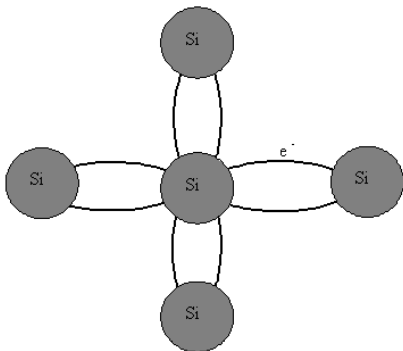
The Silicon Atom

A **covalent bond** can be formed between two atoms which have only one electron in an outer orbit or energy level. In this case the individual electrons from the separate atoms at the same energy level orbit both atoms jointly as shown in figures.

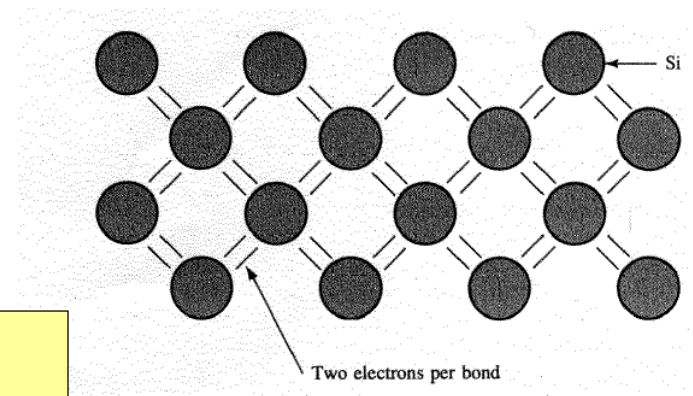


Both atoms essentially share the pair of electrons at the given energy level in the outer sub-shell, with the two electrons having opposite spins. This forms a bonding attraction between the two atoms which is not extremely strong but is nonetheless powerful and maintains a high degree of stability in the material.

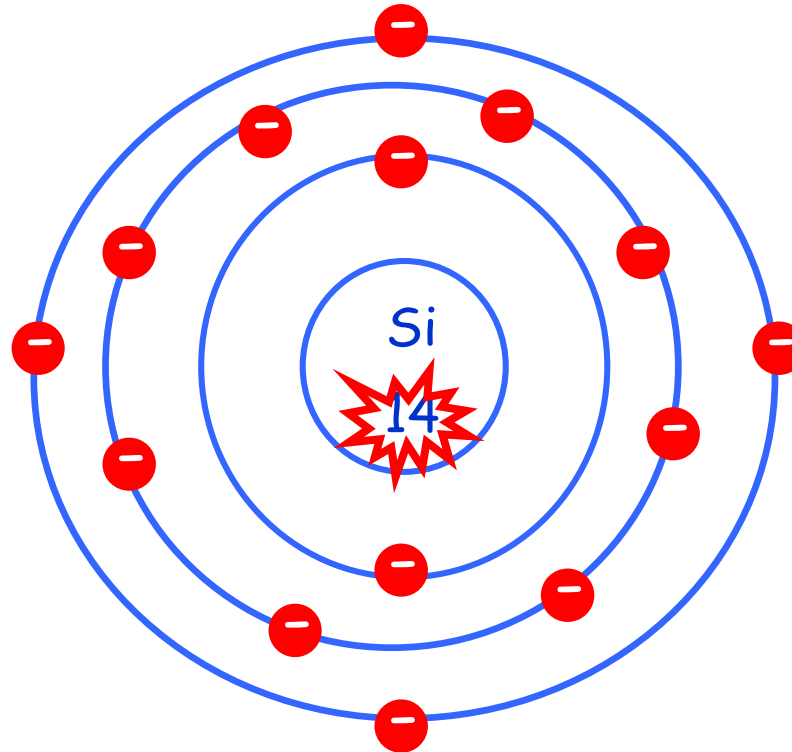
In the case of Silicon, each of the 4 outer electrons enters into a covalent bond with a neighbouring atom.



A Covalent Bond Formed by the Sharing of Electrons in an Outer Energy Level



The Silicon Atomic Structure



Silicon: our primary example and focus

Atomic no. 14

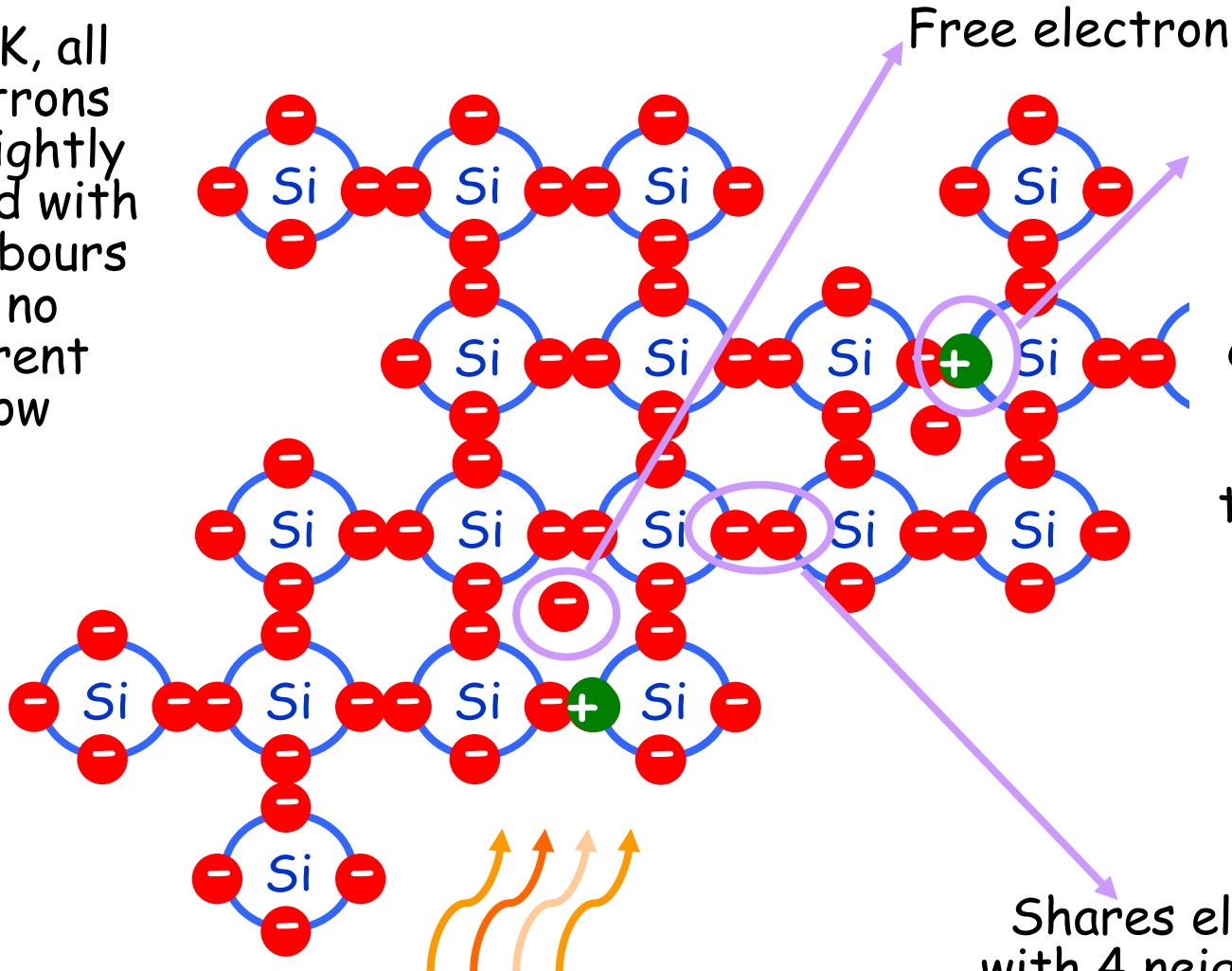
14 electrons in three shells: 2) 8) 4
i.e., 4 electrons in the outer "bonding"
shell

Silicon forms strong covalent bonds with
4 neighbors

However, like all
other elements it
would prefer to have
8 electrons in its
outer shell

Silicon Lattice Structure

At 0K, all electrons are tightly shared with neighbours
→ no current flow



Vacancy left by electron. Overall charge on silicon is zero → this "hole" must be positive

Adding heat (even to room temperature) allows some bonds to break, and electrons can flow

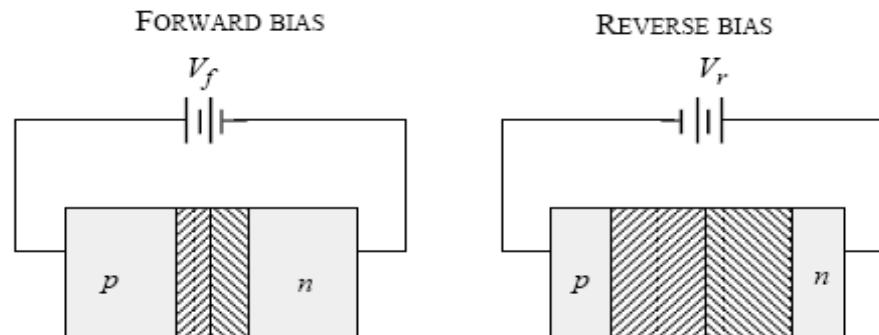
Shares electrons with 4 neighbouring atoms → 8 electrons in outer shell

What are P-type and N-type ?

- Semiconductors are classified in to P-type and N-type semiconductor
- P-type: A P-type material is one in which holes are majority carriers i.e. they are positively charged materials (++++)
- N-type: A N-type material is one in which electrons are majority charge carriers i.e. they are negatively charged materials (-----)

Forward Bias and Reverse Bias

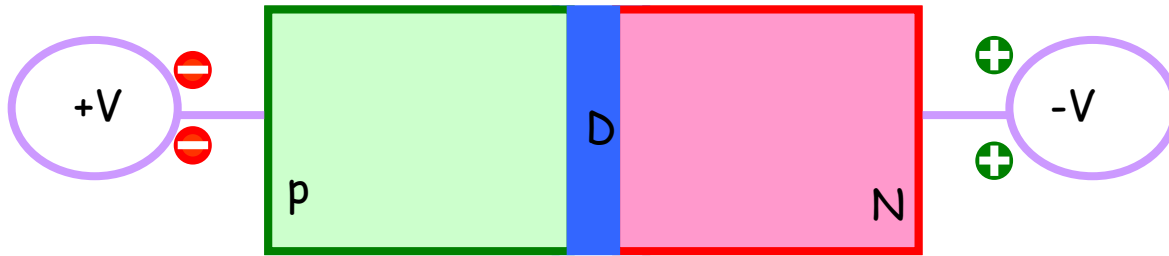
- Forward Bias : Connect positive of the Diode to positive of supply...negative of Diode to negative of supply
- Reverse Bias: Connect positive of the Diode to negative of supply...negative of diode to positive of supply.



PN Junction = Diode

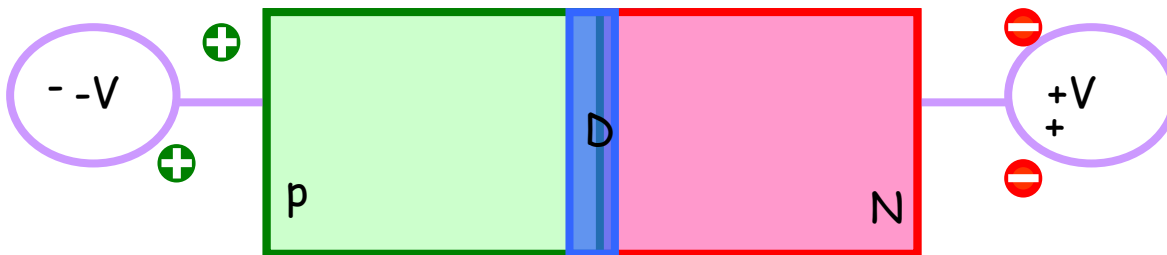
BUT – PN is no ordinary capacitor, actually a diode

Forward Bias: Shrink depletion region, current dragged through the barrier



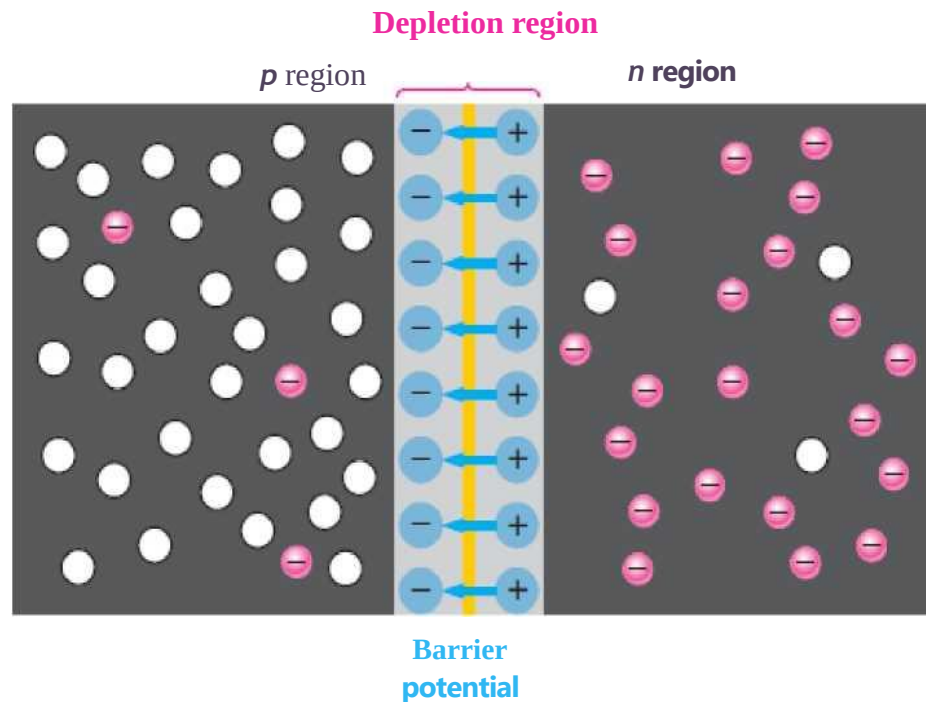
Once the difficulty of getting through the depletion region has been overcome, current can rise with applied voltage (Ohm's law)

Reverse Bias: Grow depletion region, current finds it more and more difficult to get through the barrier



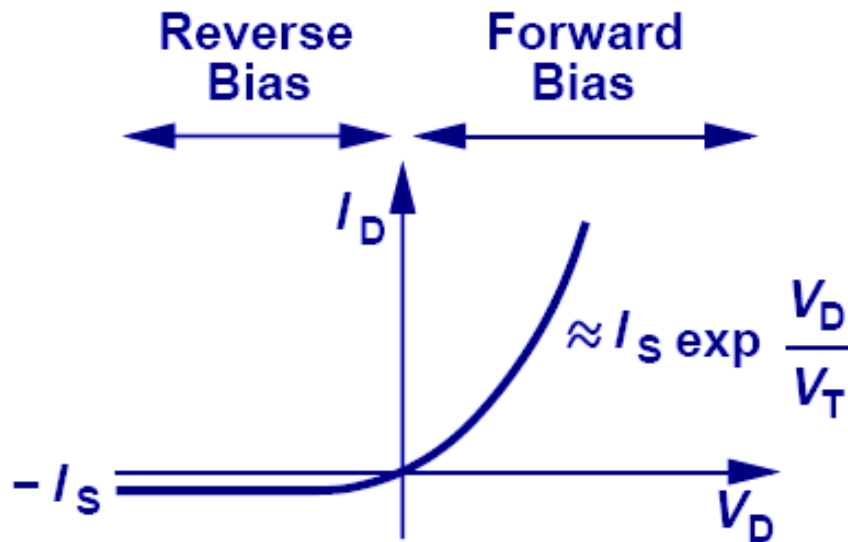
- Little current flows because barrier too high
- However increasing voltage further $\rightarrow$ high electric field
- Depletion region eventually breaks down $\rightarrow$ reverse current₁₃

These two layers of positive and negative charges form the **depletion region**, as shown in Figure below:



For every electron that diffuses across the junction and combines with a hole, a positive charge is left in the *n* region and a negative charge is created in the *p* region, forming a barrier potential. This action continues until the voltage of the barrier repels further diffusion. The blue arrows between the positive and negative charges in the depletion region represent the electric field.

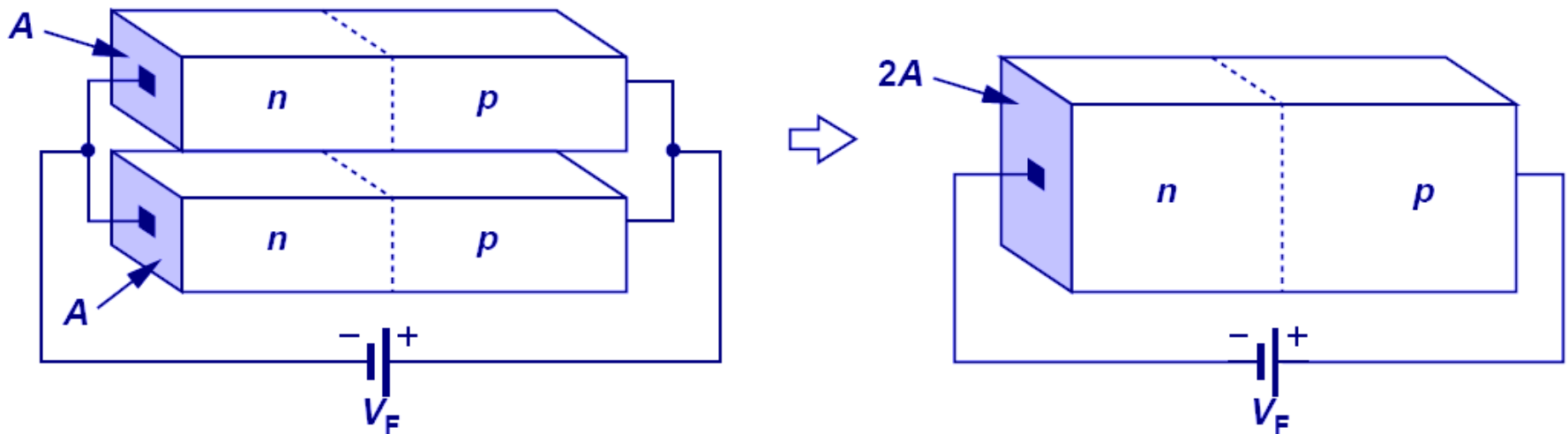
Characteristic of PN Junction



$$I_D = I_S \left(\exp \frac{V_D}{V_T} - 1 \right)$$

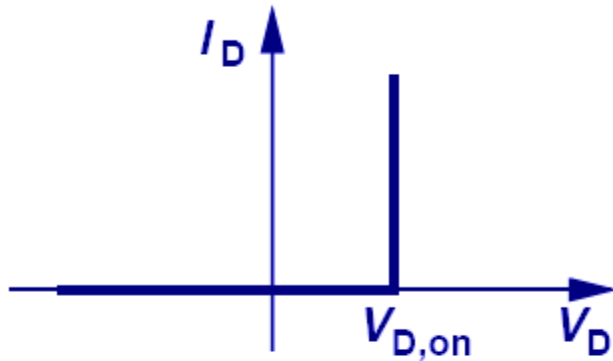
The current and voltage relationship of a PN junction is exponential in forward bias region, and relatively constant in reverse bias region.

Parallel PN Junctions

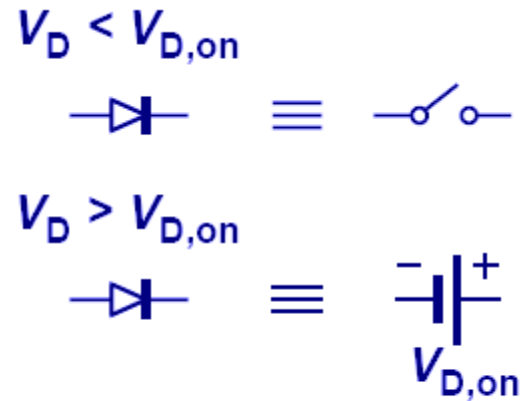


- Since junction currents are proportional to the junction's cross-section area. Two PN junctions put in parallel are effectively one PN junction with twice the cross-section area, and hence twice the current.

Constant-Voltage Diode Model



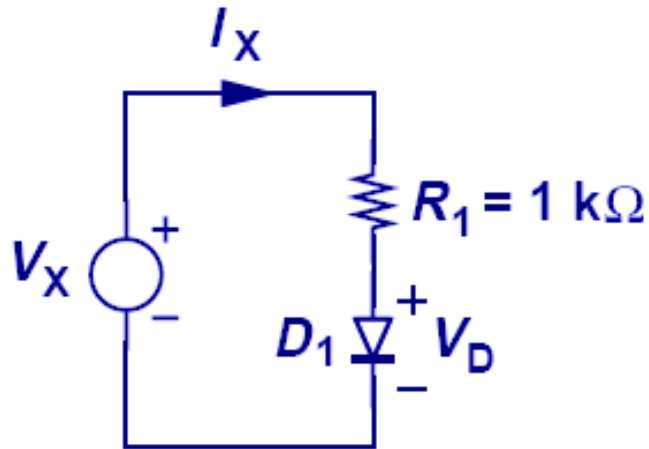
(a)



(b)

Diode operates as an open circuit if $V_D < V_{D,on}$ and a constant voltage source of $V_{D,on}$ if V_D tends to exceed $V_{D,on}$.

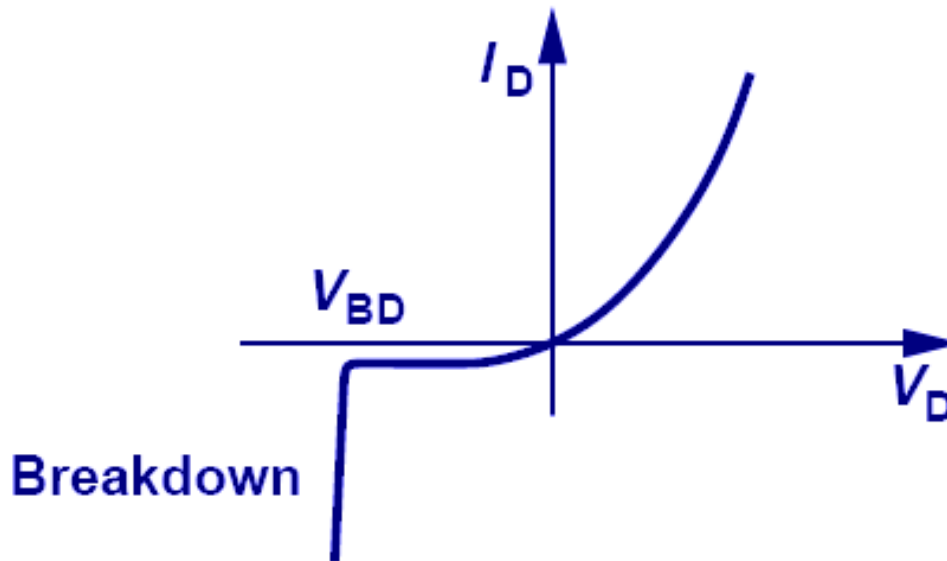
Example: Diode Calculations



$$V_X = I_X R_1 + V_D = I_X R_1 + V_T \ln \frac{I_X}{I_S}$$
$$I_X = 2.2\text{mA} \quad \text{for} \quad V_X = 3\text{V}$$
$$I_X = 0.2\text{mA} \quad \text{for} \quad V_X = 1\text{V}$$

- This example shows the simplicity provided by a constant-voltage model over an exponential model.
- For an exponential model, iterative method is needed to solve for current, whereas constant-voltage model requires only linear equations.

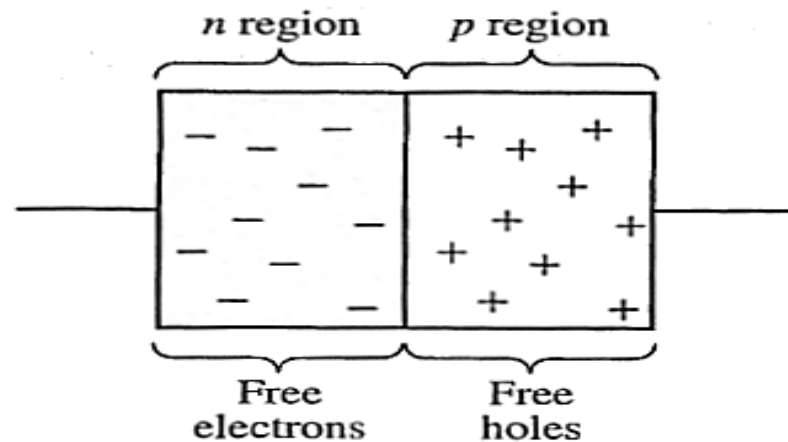
Reverse Breakdown



When a large reverse bias voltage is applied, breakdown occurs and an enormous current flows through the diode.

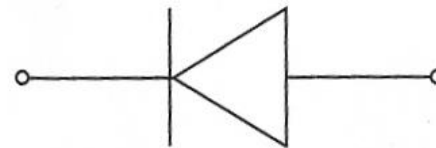
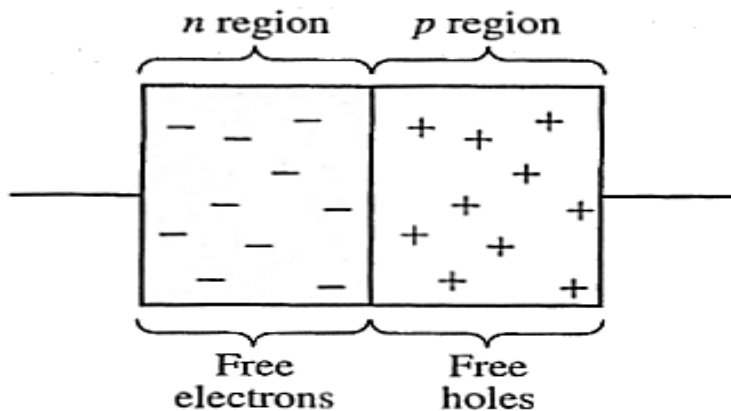
Diodes

Electronic devices created by bringing together a p -type and n -type region within the same semiconductor lattice. Used for rectifiers, LED etc



Diodes

It is represented by the following symbol, where the arrow indicates the direction of positive current flow.

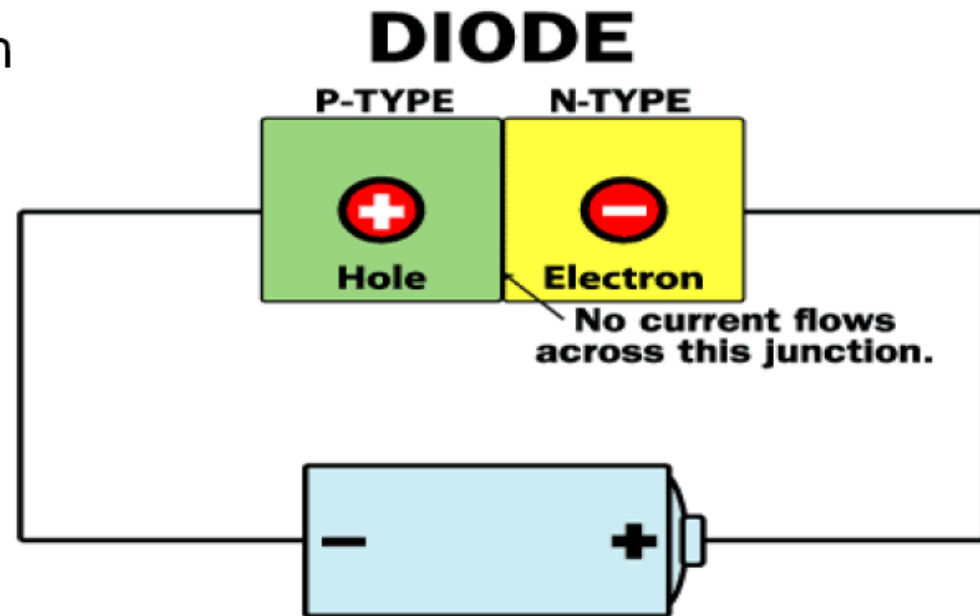


The Diode

- The diode is a two terminal semiconductor device that allows current to flow in only one direction.
- It is constructed of a P and an N junction connected together.

Diode Operation

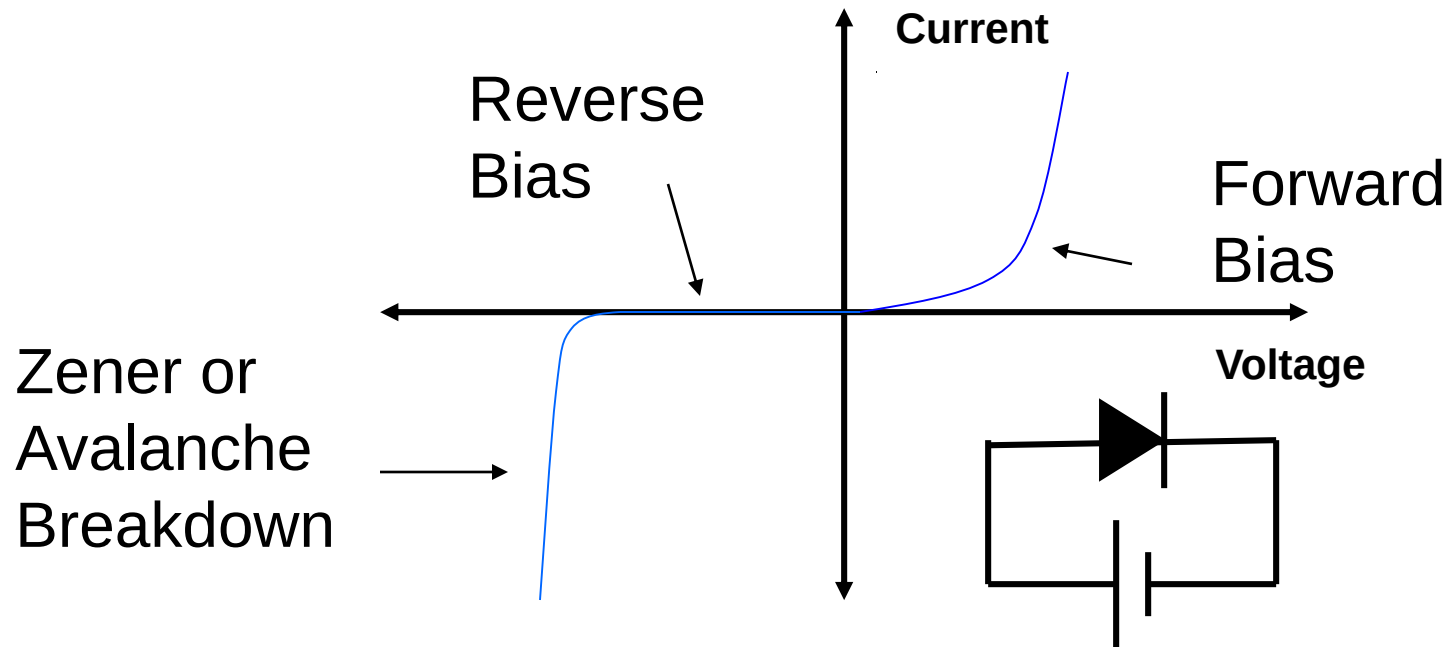
- No current flows because the holes and electrons are moving in the wrong direction.
- If you flip the battery around, the electrons are repelled by the negative terminal and the holes are repelled by the positive terminal allowing current to flow.



Characteristics of Diode

- Diode always conducts in one direction.
- Diodes always conduct current when “Forward Biased” (Zero resistance)
- Diodes do not conduct when Reverse Biased (Infinite resistance)

Working of a PN junction

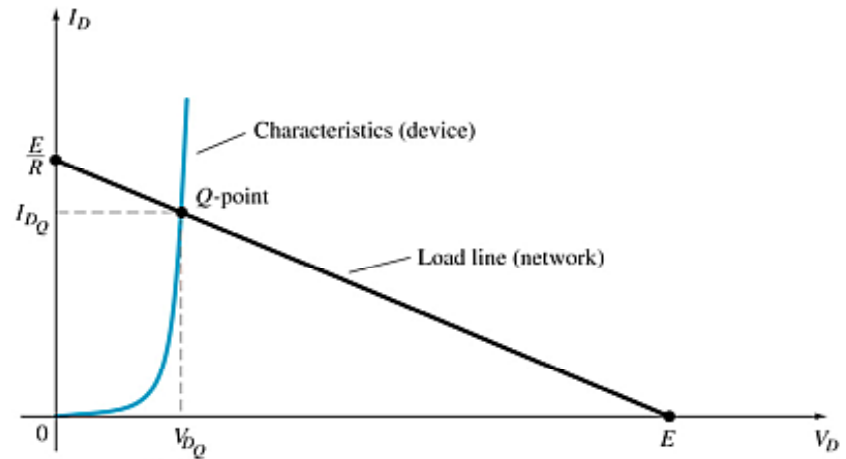


- PN junction diode acts as a *rectifier*.
- Certain current flows in forward bias mode.
- Negligible current flows in reverse bias mode until zener or avalanche breakdown happens.

Load-Line Analysis

The load line plots all possible combinations of diode current (I_D) and voltage (V_D) for a given circuit. The maximum I_D equals E/R , and the maximum V_D equals E .

The point where the load line and the characteristic curve intersect is the Q-point, which identifies I_D and V_D for a particular diode in a given circuit.



Series Diode Configurations

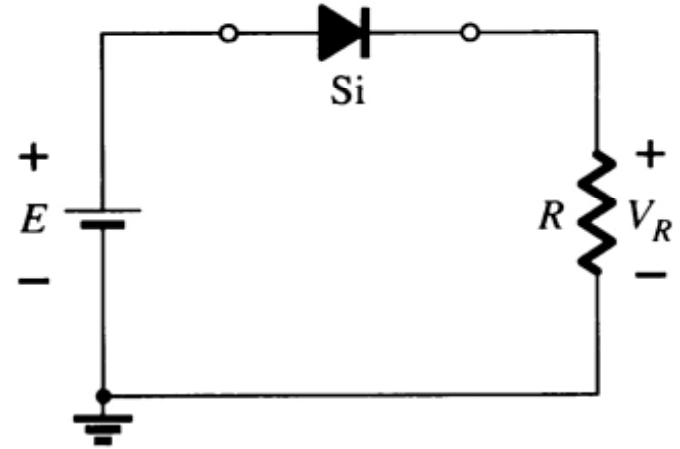
Forward Bias

Constants

- **Silicon Diode:** $V_D = 0.7 \text{ V}$
- **Germanium Diode:** $V_D = 0.3 \text{ V}$

Analysis (for silicon)

- $V_D = 0.7 \text{ V}$ (or $V_D = E$ if $E < 0.7 \text{ V}$)
- $V_R = E - V_D$
- $I_D = I_R = I_T = V_R / R$



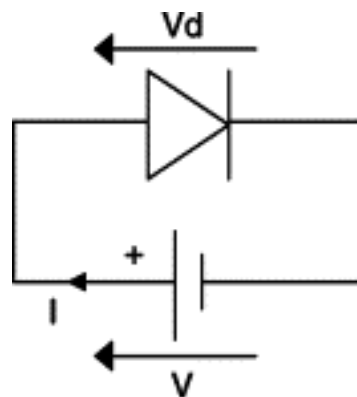
Series Diode Configurations

Reverse Bias

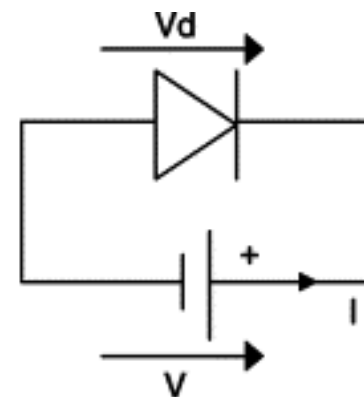
Diodes ideally behave as open circuits

Analysis

- $V_D = E$
- $V_R = 0 \text{ V}$
- $I_D = 0 \text{ A}$

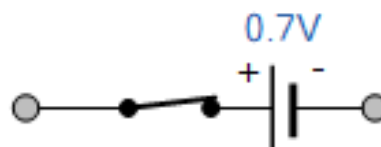


Forward biased



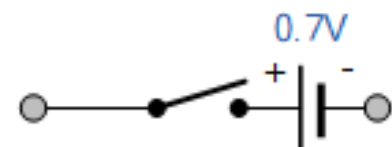
Reverse biased

Forward Biased



Forward Bias
(switch closed)

Reversed Biased

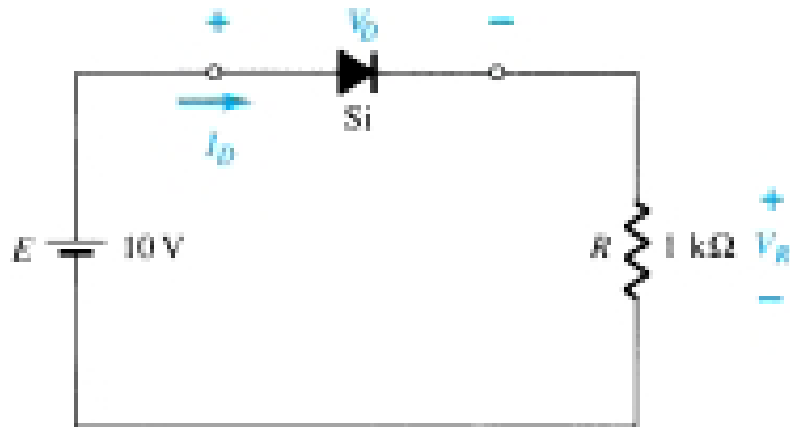


Reverse Bias
(switch open)

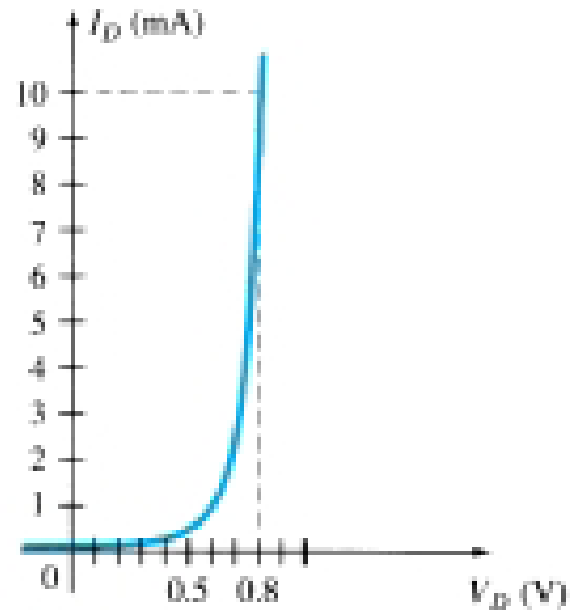
EXAMPLE

For the series diode determine

(a) V_{DQ} and I_{DQ} .



(a)



(b)

$$(a) \ I_D = \left. \frac{E}{R} \right|_{V_D=0 \text{ V}} = \frac{10 \text{ V}}{(1 \text{ k}\Omega)} = 10 \text{ mA}$$

$$V_D = E|_{I_D=0 \text{ A}} = 10 \text{ V}$$

The intersection between the load line and the characteristic curve defines the Q -point as

$$V_{D_Q} \cong \mathbf{0.78 \text{ V}}$$

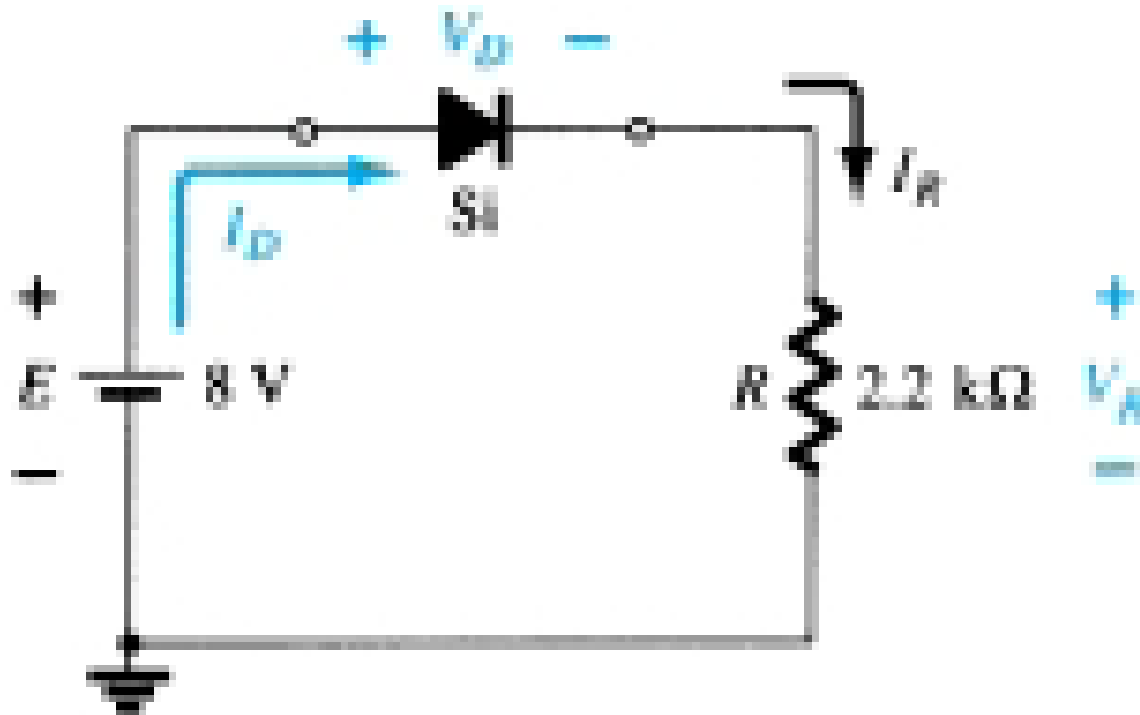
$$I_{D_Q} \cong \mathbf{9.25 \text{ mA}}$$

$$(b) \ V_R = I_R R = I_{D_Q} R = (9.25 \text{ mA})(1 \text{ k}\Omega) = \mathbf{9.25 \text{ V}}$$

$$\text{or } V_R = E - V_D = 10 \text{ V} - 0.78 \text{ V} = \mathbf{9.22 \text{ V}}$$

EXAMPLE

For the series diode configuration, determine V_D , V_R , and I_D .



- Since the applied voltage establishes a current in the clockwise direction to match the arrow of the symbol and the diode is in the “on” state,

$$V_D = \mathbf{0.7\ V}$$

$$V_R = E - V_D = 8\ \text{V} - 0.7\ \text{V} = \mathbf{7.3\ V}$$

$$I_D = I_R = \frac{V_R}{R} = \frac{7.3\ \text{V}}{2.2\ \text{k}\Omega} \cong \mathbf{3.32\ \text{mA}}$$